F.1 Simple Interest (Skipping Discounts)

Some terms

Principle or present value: the amount of money we have now

e.g. bank account → initial amount/deposit.
loan : initial loan amount

interest rate amount of interest charged expressed as % per year.

Simple interest basic interest on principle
(not compounded → interest doesn't earn interest).

future value / future amount amount after we've earned interest for some amount of time.

Simple Interest

You invest $100 for 5 years at a simple interest rate of 3%. How much interest do you have at the end of each year?

end of year 1: $100(.03) = \boxed{3}$

2: $\underbrace{100(.03)}_{\text{first year}} + \underbrace{100(.03)}_{\text{second year}} = \boxed{6}$

3: $100(.03) + 100(.03) + 100(.03) = \boxed{9}$

4: $\underbrace{9}_{\text{first 3}} + \underbrace{100(.03)}_{\text{4th year}} = \boxed{12}$

5: $5(100)(.03) = 5 \cdot 3 = \boxed{15}$

How much $ after 5 years

Future value

$F = P + I$

$= 100 + 15$

$= \boxed{115}$.

In general: Principle: P, time of investment: t

annual simple interest rate: r (as a decimal)

$I = P(r) \cdot t = \boxed{Prt}$

future value $F = P + I = P + Prt = \boxed{P(1 + rt)}$

$$I = Prt \qquad F = P(1+rt)$$

e.g. You borrow $1200 for 18 months at a simple interest rate of 5%. How much interest is owed?

$P = 1200$, $t = 1.5$ years, $r = 0.05$

$$I = 1200(.05)(1.5) = \$90$$

e.g. How much money should you put into an account that earns 3% simple interest if you wont $1000 after 10 years.

$F = 1000$, $r = 0.03$, $t = 10$

$$1000 = P(1 + .03(10))$$

$$\frac{1000}{1.3} = \frac{P(1.3)}{1.3}$$

$$P = \frac{1000}{1.3} = \$769.23$$

$$\boxed{I = Prt \qquad F = P(1+rt)}$$

e.g. We have $1000. We want $1,000,000 in 100 years. What simple interest rate do we need?

$F = 1{,}000{,}000$ $\quad t = 100$

$P = 1000$

$$\frac{1{,}000{,}000}{1000} = \frac{1000 \cdot (1 + 100r)}{1000}$$

$$1000 = 1 + 100r$$

$$-1 \qquad -1$$

$$999 = 100r$$

$$\frac{999}{100} = \frac{100r}{100}$$

$r = 9.99$ ⇝ 999% interest rate

e.g. What interest rate do we need for our money to double in 10 years?

$P = ??$

$F = 2P$

$t = 10$

$$\frac{2P}{P} = \frac{P(1 + 10r)}{P}$$

$$2 = (1 + 10r)$$

$$-1 \quad -1$$

$1 = 10r$

$r = 0.1$ ⇝ 10% interest rate